🤖 Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the simulation of human intelligence in machines. AI systems can perform tasks that normally require human intelligence such as learning, reasoning and problem solving.

📝 What is Artificial Intelligence?

AI is a technology that allows computers and machines to perform intelligent activities. It uses algorithms and data to make decisions and improve performance.

⭐ Applications of AI

- Virtual Assistants

- Self Driving Cars
- Medical Diagnosis
- Recommendation Systems
- Robotics
- Fraud Detection

🧠 Types of AI

- **Narrow AI:** Designed for specific tasks.
- **General AI:** Human-level intelligence concept.
- **Super AI:** Future concept beyond human intelligence.

🔍 Search Algorithms

Search algorithms help AI agents find solutions by exploring possible states.

- Breadth First Search (BFS)

- Depth First Search (DFS)
- A* Algorithm
- Greedy Search
- Uniform Cost Search

🌳 Intelligent Agents

An intelligent agent is a system that perceives its environment and takes actions to achieve goals.

- Simple Reflex Agent
- Model Based Agent
- Goal Based Agent
- Utility Based Agent

📚 Knowledge Representation

Knowledge representation is the method of storing information so that an AI system can understand and use it.

- Logic
- Rules
- Semantic Networks
- Frames

⚙ Expert Systems

Expert systems are AI programs that use knowledge bases and rules to solve problems like human experts.

🤖 Machine Learning Basics

Machine Learning is a part of AI that allows computers to learn from data without being explicitly programmed.

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

🎯 AI Interview Questions

Q1: What is AI?

AI is the ability of machines to perform tasks requiring human intelligence.

Q2: What is an Intelligent Agent?

An agent perceives the environment and takes actions to achieve goals.

Q3: What is Machine Learning?

Machine Learning enables computers to learn from data.

Q4: Name AI Search Algorithms.

BFS, DFS, A*, Greedy Search and Uniform Cost Search.

⚡ Final Quick Revision

Full Form: Artificial Intelligence

Field: Computer Science

Main Concepts: Agents, Search, Knowledge, ML

Uses: Automation and Decision Making